Notes: Gorton & Metrick (JFE, 2012)
Securitized Banking and the Run on Repo

Contents

1 Big picture

- **Question.** Why did subprime deterioration become a systemic crisis rather than a contained mortgage-market loss?
- **Main answer.** The crisis was a run on repo in the securitized-banking system.
- **Mechanism.** Uncertainty about bank solvency and collateral liquidity lowered collateral values, raised repo rates, and increased haircuts.
- **Empirical strategy.** The paper compares ABX, a proxy for subprime fundamentals, with LIB-OIS, a proxy for interbank counterparty risk.
- **Main finding.** LIB-OIS is strongly associated with spreads on non-subprime securitized bonds and with repo rates. Haircuts are more associated with expected collateral volatility.

2 Incremental contribution

- **Conceptual reframing.** The paper reframes the 2007–2008 crisis as a modern bank run in repo markets.
- **New architecture.** It formalizes the concept of securitized banking: securitization plus repo funding.
- **Run mechanism.** It identifies repo haircuts as the modern analogue of deposit withdrawals.
- **New data.** It uses novel data on securitized-bond spreads and repo rates/haircuts.
- **Contagion evidence.** It shows that stress moved from subprime assets to non-subprime collateral through counterparty and collateral-risk channels.

3 Data and measurement

3.1 State variables

ABX proxies for subprime mortgage fundamentals. LIB-OIS proxies for interbank counterparty risk. The empirical distinction between ABX and LIB-OIS is central to the paper.

Interpretation. ABX is the mortgage shock; LIB-OIS is the banking-system stress variable.

3.2 Credit spreads, repo spreads, and haircuts

The authors use data on 392 securitized bonds and related assets. They also study repo spreads over OIS and repo haircuts across collateral classes.

Interpretation. Haircuts are the key run variable: an increase in haircut is equivalent to a withdrawal of funding.

4 Models and empirical specifications

4.1 Balance-sheet intuition

$$\text{Cash raised} = (1 - h) \times \text{collateral value.}$$

Interpretation. If repo collateral equals $10 trillion and haircuts rise by 20 percentage points, funding capacity falls by about $2 trillion.

4.2 Credit-spread regression

$$\Delta S_{i,t} = \alpha_i + \beta \Delta LIBOIS_t + \gamma \Delta ABX_t + \delta X_t + \varepsilon_{i,t}.$$

Interpretation. The regression asks whether non-subprime spreads move with banking-system counterparty risk after controlling for subprime fundamentals.

4.3 Repo-rate and haircut regressions

$$\Delta R_{j,t} = a + b_1 \Delta ABX_t + b_2 \Delta LIBOIS_t + b_3 \Delta X_t + b_4 \Delta VOL_{j,t} + e_{j,t},$$
$$\Delta H_{j,t} = a + b_1 \Delta ABX_t + b_2 \Delta LIBOIS_t + b_3 \Delta X_t + b_4 \Delta VOL_{j,t} + e_{j,t}.$$

Interpretation. Repo rates price counterparty risk; haircuts ration lending against risky or illiquid collateral.

5 Identification and empirical design

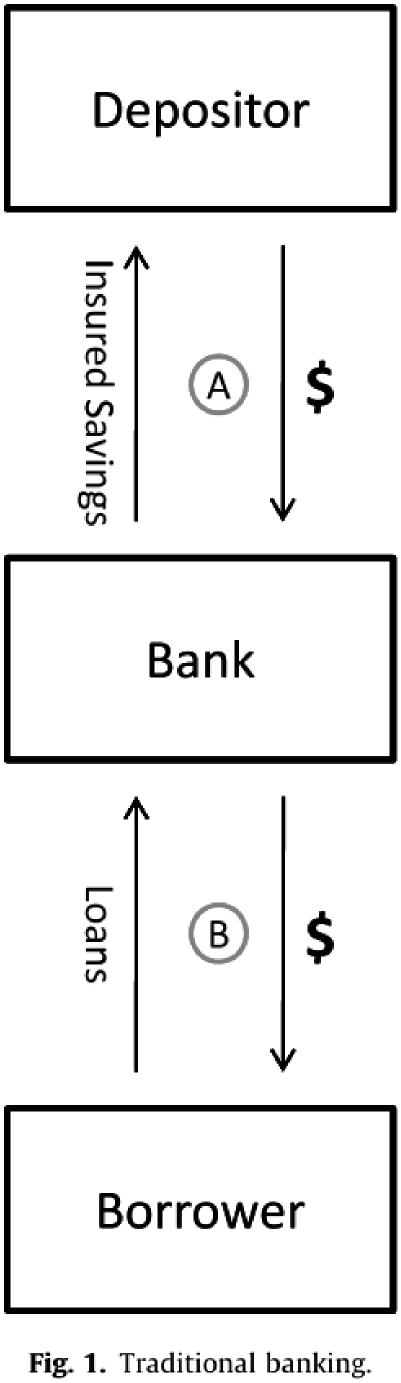
This is a mechanism and measurement paper rather than a natural experiment. Its credibility comes from separating subprime fundamentals from interbank counterparty risk and from showing that non-subprime collateral classes move with LIB-OIS.

6 Tables and figures

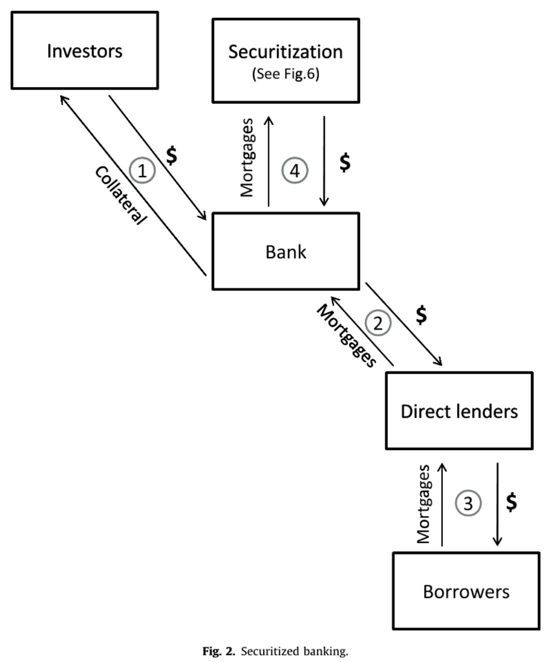
6.1 Figure 1: Traditional Banking and Figure 2: Securitized Banking

Read: Figure 1 shows insured deposits financing loans held on the bank balance sheet. Figure 2 shows repo investors, collateral, securitization, direct lenders, and borrowers.

Economic message. The fragile liability in modern banking is repo rather than retail deposits.



(a) Traditional banking

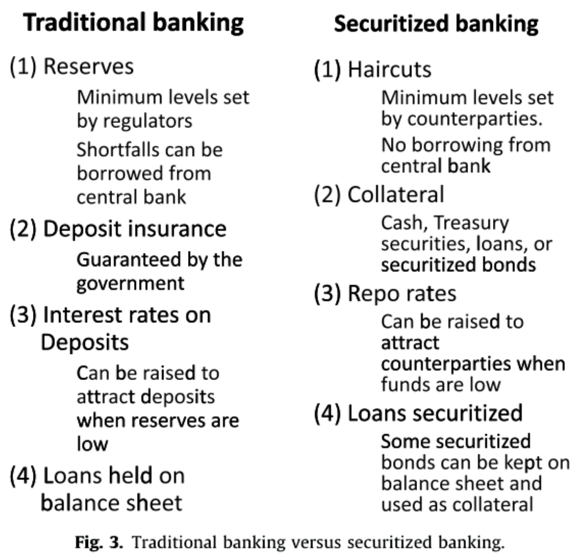


(b) Securitized banking

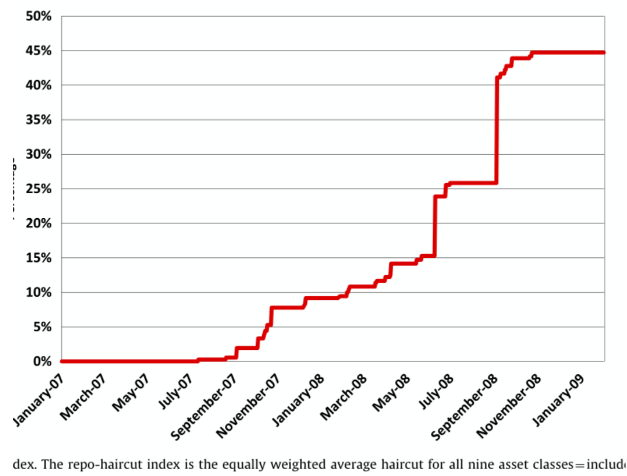
6.2 Figure 3: Traditional Banking versus Securitized Banking and Figure 4: Repo-Haircut Index

Read: Figure 3 maps reserves to haircuts, deposit insurance to collateral, and deposit rates to repo rates. Figure 4 shows the haircut index rising toward 50 percent.

Economic message. Rising haircuts are the run on repo.



(a) Traditional versus securitized banking



(b) Repo-haircut index

6.3 Figure 5: Issuance in US Capital Markets and Figure 6: Securitization

Read: Figure 5 shows the scale of securitization. Figure 6 shows how loans, SPVs, tranches, CDOs, and repo collateral fit together.

Economic message. Securitization manufactured collateral for repo-funded banking.

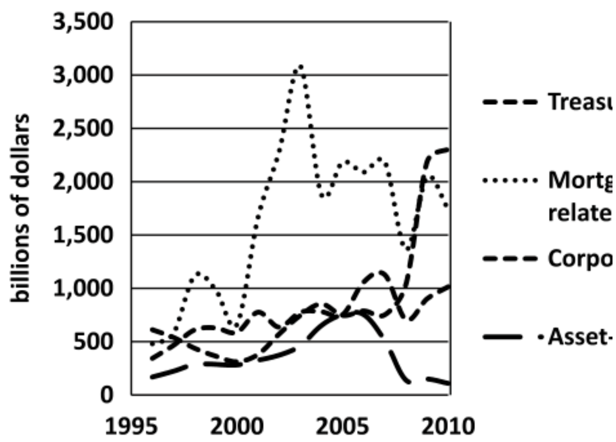


Fig. 5. Issuance in US capital markets (billions of dollars). Dat
US Department of Treasury, federal agencies, Thomson Finan
MBS & ABS, and Bloomberg.

(a) Issuance in US capital markets

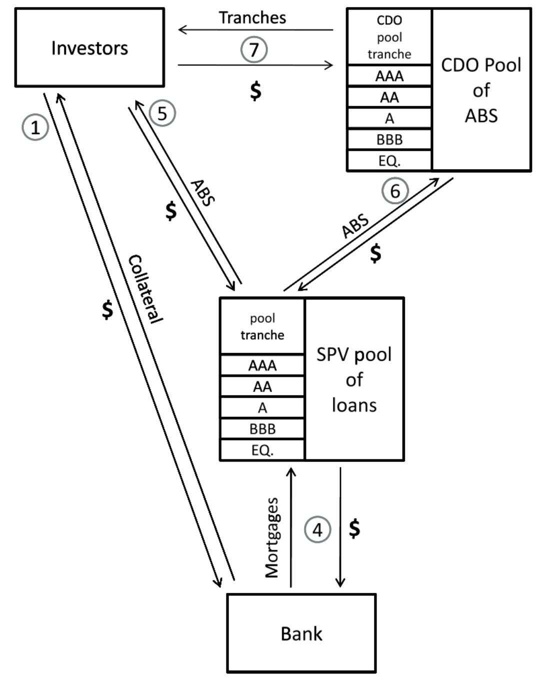


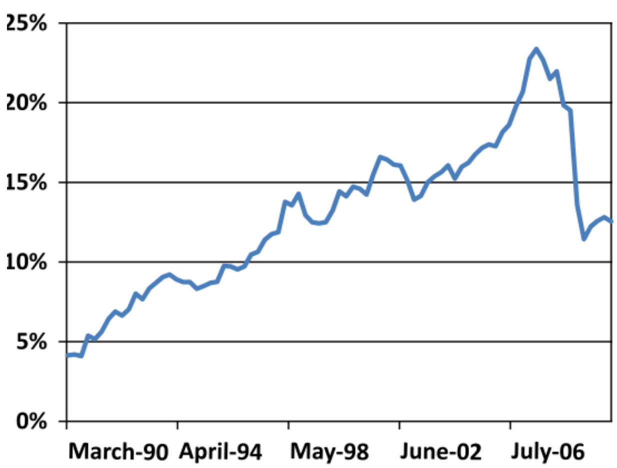
Fig. 6. Securitization.

(b) Securitization

6.4 Figure 7: Broker-Dealer Assets and Table 1: Broker-Dealer Repo Financing

Read: Figure 7 shows broker-dealer assets rising relative to banks. Table 1 documents large repo-financed dealer positions.

Economic message. The repo-funded system was large enough to be systemic.



ig. 7. Ratio of broker–dealers' total assets to banks' total assets. Data

(a) Broker-dealer assets to bank assets

Table 1
Broker–dealer repo financing (billions of dollars).

	Goldman Sachs	Morgan Stanley	Lehman Brothers	Merrill Lynch	Bear Stearns	Total
Panel A: Mid-2008						
Financial instruments owned	411,194	390,393	269,409	288,925	141,104	1,501,025
Of which pledged (and can be repledged)	37,383	140,000	43,031	27,512	22,903	270,829
Of which pledged (and cannot be repledged)	120,980	54,492	80,000	53,025	54,000	362,497
Of which not pledged at all	252,831	195,901	146,378	208,388	64,201	867,699
% own financial instruments pledged	38.5	49.8	45.7	27.9	54.5	42.2

Panel B: End of 2006								
Financial instruments owned	334,561	380,853	226,596	312,187	125,168	1,379,365	287,258	203,848
Of which pledged (and can be repledged)	35,998	125,000	42,600	58,966	15,967	278,531	58,966	58,966
Of which pledged (and cannot be repledged)	134,310	63,903	75,000	68,594	41,000	382,807	68,594	68,594
Of which not pledged at all	164,253	191,950	108,996	184,627	68,201	718,027	159,698	76,288
Own financial instruments pledged	50.9	49.6	51.9	40.9	45.5	47.9	44.4	62.6

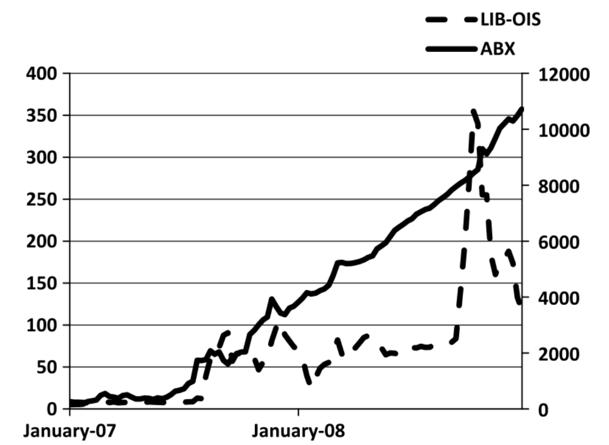
Merrill Lynch (ML) did not list Financial instruments owned. Instead, ML listed Trading assets, Investment securities, and Securities received as collateral. In the column ML, all three asset categories were used. In the column marked ML**, only trading assets and investment securities are included. In the case of ML**, only Trading assets are used. Data are from Company 10-Qs. See King (2008) and Singh and Aitken (2009).

(b) Broker-dealer repo financing

6.5 Figure 8: ABX versus LIB-OIS and Figure 9: LIB-OIS and Non-Subprime Spreads

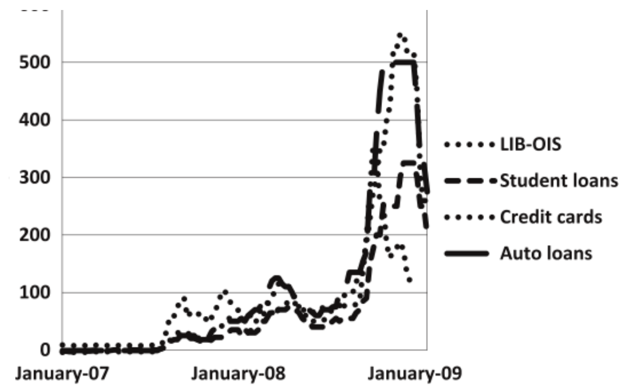
Read: ABX deteriorates steadily, while LIB-OIS spikes in crisis episodes. Non-subprime spreads co-move with LIB-OIS.

Economic message. Contagion operated through counterparty risk and collateral liquidity, not only mortgage fundamentals.



ecurity index versus LIB-OIS (spread between the London Interbank Offered Rate and the rate on an overnigl
he.

(a) ABX versus LIB-OIS



ʒ. 9. LIB-OIS (spread between the London interbank offered rate and e rate on an overnight index swap) and non-subprime-related asset isses. LIB-OIS is shown with the spreads on AAA-rated asset-backed curities: student loans, credit cards, and auto loans. The scale is in sis points.

(b) LIB-OIS and non-subprime spreads

6.6 Table 2: Summary Statistics

Read: Table 2 documents the spread and haircut behavior across subprime, non-subprime, repo, and control variables.

Economic message. Stress spreads far beyond subprime assets.

Table 2
Summary statistics.
This table reports the summary statistics for the state variable, credit spreads, repo spreads and control variables used in this paper. For each series we show summary statistics for the whole period and four subperiods. Panel A shows statistics for the state variable LIB-OIS and the credit spreads of three categories of subprime related assets. Panels B and Panel C shows statistics for the credit spreads of US and non-US non-subprime asset classes, respectively. Panel D reports the statistics for the spreads between three-month repo rates and OIS. Panel E shows statistics for the five control variables used in the regression analysis. All variables given in this table are defined in Appendix A. All spreads are measured in basis points, with spread computations explained in Appendix B.

Series	Periods	Mean	Median	Std. Err.	Max	Min
Panel A: State variable and subprime related assets class						
State variable						
LIB-OIS spread	Whole period	72.43	66.44	67.57	354.20	7.45
	First half of 2007	7.97	7.88	0.42	9.15	7.45
	Second half of 2007	58.71	60.78	28.64	104.73	7.70
	All of 2007	33.34	8.50	32.53	104.73	7.45
	All of 2008	108.10	77.20	71.61	354.20	24.33
Credit spreads of sub-prime related assets						
HEL BBB	Whole period	714.25	425.00	545.26	1800.00	170.00
	First half of 2007	273.27	250.00	93.54	425.00	170.00
	Second half of 2007	1113.46	1000.00	441.03	1800.00	350.00
	All of 2007	693.37	425.00	528.75	1800.00	170.00
Mezzanine CDO BBB	Whole period	2861.93	2969.42	2023.57	8421.76	365.00
	First half of 2007	627.50	650.00	152.50	950.00	365.00
	Second half of 2007	2178.47	1940.75	659.41	3063.16	1100.00
	All of 2007	1402.99	1025.00	915.27	3063.16	365.00
	All of 2008	4858.38	4687.97	1268.88	8421.76	3016.83
HELOC AAA	Whole Period	121.60	18.00	157.64	500.00	14.00
	First half of 2007	15.35	15.00	1.23	18.00	14.00
	Second half of 2007	213.31	195.00	159.59	500.00	15.00
	All of 2007	114.33	18.00	149.92	500.00	14.00
CDS spreads of subprime mortgage lenders						
Countrywide	Whole period	275.86	220.02	253.96	1185.84	22.52
	First half of 2007	51.10	50.79	17.64	77.57	22.52
	Second half of 2007	445.43	282.34	338.99	1088.00	62.53
	All of 2007	248.26	71.09	310.03	1088.00	22.52
	All of 2008	301.04	260.00	188.39	1185.84	95.67
WAMU	Whole period	413.57	318.33	643.09	4352.43	19.61
	First half of 2007	35.39	33.68	11.11	57.38	19.61
	Second half of 2007	189.27	122.03	133.36	438.47	42.55
	All of 2007	112.33	50.32	121.71	438.47	19.61
	All of 2008	815.23	438.90	818.44	4352.43	255.00
CDS spreads of monoline insurers						
MBIA	Whole period	911.33	391.88	1116.88	4153.55	13.46
	First half of 2007	24.20	22.66	9.34	42.20	13.46
	Second half of 2007	164.92	121.69	89.44	322.96	45.18
	All of 2007	94.56	43.69	94.93	322.96	13.46
	All of 2008	1656.46	1391.26	1101.33	4153.55	372.55
MGIC	Whole period	520.96	453.46	406.35	1411.73	27.48
	First half of 2007	51.42	52.40	17.82	81.32	27.48
	Second half of 2007	270.97	245.95	138.90	528.53	71.21
	All of 2007	161.20	79.09	147.99	528.53	27.48
	All of 2008	849.17	885.61	262.06	1411.73	320.00
Radian	Whole period	1072.85	809.83	956.80	3164.27	28.53
	First half of 2007	56.54	55.49	18.12	93.39	28.53
	Second half of 2007	536.20	559.22	261.37	1000.32	73.26
	All of 2007	296.37	86.62	303.80	1000.32	28.53
	All of 2008	1781.22	1938.22	781.58	3164.27	704.70

Panel B: US Non-subprime asset classes Series	Periods	Mean	Median	Std. Err.	Max	Min
Auto AAA	Whole period	109.42	55.00	153.65	500.00	-1.00
	First half of 2007	-1.00	-1.00	0.00	-1.00	-1.00
	Second half of 2007	23.38	20.00	16.11	50.00	-1.00
	All of 2007	11.19	-1.00	16.70	50.00	-1.00
	All of 2008	199.04	110.00	167.75	500.00	50.00
Cards AAA	Whole period	101.92	55.00	148.93	550.00	-4.00
	First half of 2007	-3.31	-3.00	0.47	-3.00	-4.00
	Second half of 2007	24.88	23.00	17.73	55.00	-3.00
	All of 2007	10.79	-3.00	18.89	55.00	-4.00
	All of 2008	185.05	100.00	166.45	550.00	50.00

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Table 2 (*continued*)

Panel B: US Non-subprime asset classes Series	Periods	Mean	Median	Std. Err.	Max	Min
	First half of 2007	-1.12	-1.00	1.18	0.00	-3.00
	Second half of 2007	17.92	18.00	10.28	35.00	0.00
	All of 2007	8.40	0.00	12.03	35.00	-3.00
	All of 2008	116.84	65.00	99.71	325.00	25.00
CMBS AAA	Whole period	241.16	123.00	313.92	1350.00	22.00
	First half of 2007	26.27	26.50	2.79	31.00	22.00
	Second half of 2007	65.88	62.00	21.83	105.00	31.50
	All of 2007	46.08	31.25	25.25	105.00	22.00
	All of 2008	419.12	250.00	349.14	1350.00	83.00
HG SF CDO AAA	Whole period	482.24	428.32	424.87	1463.10	23.00
	First half of 2007	23.73	23.00	2.68	35.00	23.00
	Second half of 2007	177.88	108.09	135.50	450.90	42.00
	All of 2007	100.81	38.50	122.72	450.90	23.00
	All of 2008	830.22	849.77	275.79	1463.10	391.43
Mezzanine SF CDO AAA	Whole period	1094.17	1084.79	873.46	2712.79	32.00
	First half of 2007	54.38	55.00	20.99	105.00	32.00
	Second half of 2007	567.96	444.69	380.54	1084.79	115.00
	All of 2007	311.17	110.00	372.07	1084.79	32.00
	All of 2008	1808.49	1742.65	506.59	2712.79	1006.25
Panel C: Non-US non-subprime asset classes						
Australia RMBS AAA	Whole period	145.60	90.00	151.22	650.00	5.00
	First half of 2007	6.23	7.00	0.91	7.00	5.00
	Second half of 2007	48.15	50.00	24.66	90.00	5.00
	All of 2007	27.19	7.00	27.32	90.00	5.00
	All of 2008	257.55	200.00	133.86	650.00	90.00
UK RMBS AAA	Whole period	124.53	80.00	131.25	440.00	4.00
	First half of 2007	4.96	5.00	0.72	6.00	4.00
	Second half of 2007	45.15	50.00	22.23	80.00	5.00
	All of 2007	25.06	6.00	25.58	80.00	4.00
	All of 2008	218.58	180.00	121.08	440.00	80.00
Dutch RMBS AAA	Whole Period	56.56	50.00	42.40	170.00	10.00
	First half of 2007	10.69	11.00	0.74	12.00	10.00
	Second half of 2007	40.35	39.00	18.96	70.00	11.00
	All of 2007	25.52	12.00	20.02	70.00	10.00
	All of 2008	97.95	90.00	25.38	170.00	60.00
UK Cards AAA	Whole period	178.62	150.00	176.97	625.00	10.00
	First half of 2007	10.08	10.00	0.28	11.00	10.00
	Second half of 2007	44.38	40.00	25.94	90.00	11.00
	All of 2007	32.95	27.00	26.67	90.00	10.00
	All of 2008	281.91	215.00	165.05	625.00	90.00
European Consumer Receivable AAA	Whole period	206.25	200.00	176.36	600.00	15.00
	First half of 2007	15.29	15.00	0.76	17.00	15.00
	Second half of 2007	54.92	55.00	25.05	95.00	18.00
	All of 2007	46.52	50.00	27.59	95.00	15.00
	All of 2008	302.09	235.00	157.19	600.00	95.00
European Auto AAA	Whole period	120.57	85.00	119.56	450.00	7.00
	First half of 2007	7.14	7.00	0.38	8.00	7.00
	Second half of 2007	37.31	35.00	18.62	65.00	8.00
	All of 2007	30.91	30.00	20.68	65.00	7.00
	All of 2008	174.36	115.00	122.14	450.00	60.00

Panel D: Repo rate spreads (basis points; except repo haircuts) Series	Periods	Mean	Median	Std. Err.	Max	Min	Mean of haircut
BBB+/A Corporates	Whole period	86.50	82.14	83.15	429.43	0.50	0.5%
	First half of 2007	2.01	1.95	0.61	5.30	0.50	0.0%
	Second half of 2007	61.85	65.49	36.29	126.35	1.70	0.0%
	All of 2007	32.28	2.70	39.53	126.35	0.50	0.0%
	All of 2008	136.19	103.63	81.61	429.43	44.33	0.9%
AA-AAA Corporates	Whole period	77.59	74.78	78.42	409.43	-3.50	0.5%
	First half of 2007	-1.69	-2.05	1.90	10.44	-3.50	0.0%
	Second half of 2007	55.27	58.95	34.53	116.35	-2.30	0.0%
	All of 2007	27.13	-1.35	37.64	116.35	-3.50	0.0%

Table 2 (*continued*)

Panel D: Repo rate spreads (basis points; except repo haircuts) Series	Periods	Mean	Median	Std. Err.	Max	Min	Mean of haircut
	All of 2008	123.86	92.11	77.57	409.43	39.33	0.9%
A-AAA ABS-Auto/CC/SL	Whole period	105.22	94.76	101.00	479.43	1.70	5.2%
	First half of 2007	4.44	4.00	1.77	11.00	1.70	0.0%
	Second half of 2007	68.44	71.78	40.93	141.35	3.70	0.9%
	All of 2007	36.82	5.25	43.29	141.35	1.70	0.5%
	All of 2008	167.92	119.81	98.07	479.43	54.33	9.5%
AA-AAA ABS-RMBS/CMBS	Whole period	124.04	107.78	120.11	520.30	3.70	9.4%
	First half of 2007	6.41	6.00	1.76	13.00	3.70	0.0%
	Second half of 2007	76.35	81.78	43.92	151.35	5.70	1.8%
	All of 2007	41.80	7.00	46.92	151.35	3.70	0.9%
	All of 2008	199.44	145.08	117.27	520.30	64.33	17.1%
< AA ABS-RMBS/CMBS	Whole period	135.90	117.78	129.02	550.30	6.70	10.6%
	First half of 2007	9.41	9.00	1.76	16.00	6.70	0.0%
	Second half of 2007	84.55	88.20	48.62	166.35	8.70	3.7%
	All of 2007	47.43	10.00	51.08	166.35	6.70	1.9%
	All of 2008	217.01	153.95	125.56	550.30	69.33	18.6%
Unpriced ABS/MBS/All Sub-Prime	Whole period	108.94	109.69	84.64	295.38	7.70	37.3%
	First half of 2007	10.41	10.00	1.76	17.00	7.70	0.0%
	Second half of 2007	95.62	97.83	58.54	196.35	9.70	7.7%
	All of 2007	53.52	11.00	59.59	196.35	7.70	3.9%
	All of 2008	187.28	197.88	42.23	295.38	99.33	68.0%
AA-AAA CLO	Whole period	134.46	117.14	127.18	545.30	3.70	10.2%
	First half of 2007	6.41	6.00	1.76	13.00	3.70	0.0%
	Second half of 2007	85.93	92.65	51.27	171.35	5.70	1.8%
	All of 2007	46.64	7.00	53.98	171.35	3.70	0.9%
	All of 2008	214.96	148.76	121.61	545.30	79.33	18.7%
AA-AAA CDO	Whole period	130.09	124.69	107.46	380.38	4.70	30.0%
	First half of 2007	7.41	7.00	1.76	14.00	4.70	0.0%
	Second half of 2007	107.77	109.35	69.56	226.35	6.70	8.3%
	All of 2007	58.19	8.00	70.48	226.35	4.70	4.3%
	All of 2008	231.72	241.39	56.52	380.38	129.33	53.5%
Unpriced CLO/CDO	Whole period	148.32	142.60	123.54	413.75	6.70	32.4%
	First half of 2007	9.41	9.00	1.76	16.00	6.70	0.0%
	Second half of 2007	122.63	124.42	80.14	256.35	8.70	10.5%
	All of 2007	66.69	10.00	80.34	256.35	6.70	5.4%
	All of 2008	268.39	256.58	63.03	413.75	154.33	57.3%
Repo-Rate Index and Repo-Haircut Index (last column)	Whole period	151.36	130.89	152.79	688.10	3.81	15.1%
	First half of 2007	6.03	5.67	1.45	11.33	3.81	0.0%
	Second half of 2007	84.27	89.18	51.14	172.46	5.37	3.9%
	All of 2007	45.61	6.98	53.44	172.46	3.81	2.0%
	All of 2008	248.29	171.20	149.95	688.10	81.55	27.2%

Panel E: Control variables Series	Periods	Mean	Median	Std. Err.	Max	Min
ABX Index (basis points)	Whole period	4090.11	3604.00	3524.76	10,940.98	121.00
	First half of 2007	302.15	329.50	139.54	552.00	121.00
	Second half of 2007	1657.69	1752.00	894.91	3286.00	368.00
	All of 2007	979.92	481.50	933.01	3286.00	121.00
	All of 2008	6927.48	6938.67	2447.64	1094,0.98	3373.00
10Year Treasury rate	Whole period	4.07%	4.04%	0.71%	5.19%	2.08%
	First half of 2007	4.77%	4.73%	0.20%	5.19%	4.51%

6.7 Tables 3–4: Credit Spread Regressions and F-test Summary

Read: Table 3 shows LIB-OIS significance for many non-subprime spreads. Table 4 aggregates the evidence by asset class.

Economic message. These are the central contagion tests.

Table 3
Credit spreads regression results.

For each bond i, we estimate Eq. (2) using weekly data from January 4, 2007 to January 29, 2009, ΔLIB-OIS is the percentage change of the spread between the 3-month LIBOR and the Overnight Index Swap (OIS). ΔABX is the parentage change of the index at period t. ΔOIS is the Overnight Index Swap. $\Delta r-10$ is the change in yield on the 10-year Treasury, with its square given by $(\Delta r-10)^2$. ΔSlope is the change in 10-year minus 2-year Treasury yields. ΔVIX is the change in implied volatility of Standard and Poor 500, and ΔS&P is the return on S&P 500. t-statistics are given in parentheses below the coefficient estimates. The last two rows report F-statistics and p-values for the key state variables. The null hypothesis of the LIB-OIS F-test is that the sum of all coefficients of ΔLIB-OIS and its lags is zero. The null hypothesis of the ABX F-test is the sum of all coefficients of ΔABX and its lags is zero. Panel A shows the results of six US non-subprime assets and Panel B shows the results of six non-US non-subprime assets.

Panel A: US non-subprime asset classes

	Credit spreads					
	Cards	Auto	Student	CMBS	HG SF CDO	Mezz SF CDO
Intercept	0.003	0.016	−0.010	0.036	0.042	0.052
	(0.1)	(0.33)	(−0.33)	(2.41)	(2.86)	(3.96)
ΔLIB-OIS	0.341	0.079	0.461	0.025	−0.051	−0.037
	(3.24)	(0.54)	(5.23)	(0.26)	(−1.16)	(−0.94)
ΔLIB-OIS, $t-1$	0.264	0.486	0.131	0.078	−0.042	0.055
	(2.64)	(3.55)	(1.59)	(0.84)	(−1.00)	(1.45)
ΔLIB-OIS, $t-2$	0.132	0.012	0.138	−0.082	0.038	−0.081
	(1.32)	(0.08)	(1.67)	(−0.92)	(0.91)	(−2.15)
ΔLIB-OIS, $t-3$	0.027	0.170	−0.013	−0.030	−0.030	−0.004
	(0.27)	(1.25)	(−0.16)	(−0.33)	(−0.72)	(−0.1)
ΔABX	−0.141	−0.331	0.455	0.012	0.001	0.070
	(−0.66)	(−1.13)	(2.32)	(0.27)	(0.00)	(0.86)
ΔABX, $t-1$	0.079	−0.025	0.119	−0.013	0.016	0.061
	(0.36)	(−0.09)	(0.6)	(−0.32)	(0.18)	(0.74)
ΔABX, $t-2$	0.315	0.250	−0.202	−0.072	−0.020	−0.040
	(1.48)	(0.86)	(−1.06)	(−1.71)	(−0.23)	(−0.5)
ΔABX, $t-3$	−0.277	−0.351	−0.150	−0.052	0.049	−0.011
	(−1.3)	(−1.2)	(−0.69)	(−1.24)	(0.54)	(−0.14)
ΔOIS	−0.253	−0.147	−0.358	−0.096	0.156	0.106
	(−0.78)	(−0.33)	(−1.34)	(−0.69)	(1.14)	(0.85)
$\Delta r-10$	0.111	−0.092	0.059	−0.214	−0.227	−0.132
	(0.58)	(−0.36)	(0.36)	(−2.65)	(−2.87)	(−1.85)
$(\Delta r-10)^2$	0.174	0.076	0.037	−0.042	−0.094	−0.144
	(0.57)	(0.18)	(0.14)	(−0.33)	(−0.75)	(−1.26)
ΔS&P	−0.443	1.518	−0.757	−1.622	−0.580	0.592
	(−0.29)	(0.7)	(−0.57)	(−2.42)	(−0.89)	(0.99)
ΔVIX	−0.006	0.004	−0.003	0.003	0.002	0.006
	(−0.5)	(0.23)	(−0.29)	(0.56)	(0.41)	(1.26)
ΔSlope	−0.155	0.189	−0.039	0.298	0.269	0.075
	(−0.74)	(0.64)	(−0.23)	(3.29)	(3.02)	(0.92)
LIB-OIS F-test	20.16	10.26	26.13	2.99	1.38	1.08
	(< 0.01)	(< 0.01)	(< 0.01)	(0.08)	(0.24)	(0.30)
ABX F-test	0.00	0.73	0.42	0.00	0.08	0.39
	(0.95)	(0.40)	(0.52)	(0.95)	(0.78)	(0.59)

Panel B: Non-US non-subprime asset classes

	Credit spreads					
	Australia RMBS	UK RMBS	Dutch RMBS	UK cards	European consumer receivable	European auto
Intercept	0.014	0.017	0.014	0.031	0.032	0.037
	(0.52)	(0.93)	(1.14)	(2.4)	(3.21)	(2.35)
ΔLIB-OIS	0.126	0.240	0.100	0.109	0.049	0.081
	(1.57)	(4.38)	(2.71)	(2.86)	(1.76)	(1.8)
ΔLIB-OIS, $t-1$	0.575	0.237	0.071	0.136	0.021	0.019
	(7.56)	(4.57)	(2.11)	(3.72)	(0.8)	(0.45)
ΔLIB-OIS, $t-2$	−0.181	−0.051	0.115	0.019	0.033	0.104
	(−2.44)	(−1.01)	(3.42)	(0.52)	(1.26)	(2.51)
ΔLIB-OIS, $t-3$	0.138	0.067	−0.015	−0.019	0.020	0.022
	(1.86)	(1.32)	(−0.47)	(−0.55)	(0.77)	(0.53)
ΔABX	0.025	0.029	0.095	0.094	0.116	0.095
	(0.15)	(0.26)	(1.37)	(1.03)	(1.65)	(0.84)
ΔABX, $t-1$	−0.002	0.022	0.002	0.028	0.001	−0.010
	(−0.02)	(0.19)	(0.03)	(0.34)	(0.01)	(−0.09)
ΔABX, $t-2$	−0.171	−0.018	−0.037	0.011	0.037	0.044
	(−1.09)	(−0.17)	(−0.54)	(0.13)	(0.54)	(0.4)
ΔABX, $t-3$	0.173	0.072	−0.109	−0.084	−0.060	−0.265
	(1.09)	(0.66)	(−1.6)	(−1.05)	(−0.9)	(−2.47)
ΔOIS	0.034	−0.031	−0.237	−0.051	−0.054	−0.164

Table 3: Credit-spread regressions

Table 3 (*continued*)

Panel B: Non-US non-subprime asset classes

	Credit spreads					
	Australia RMBS	UK RMBS	Dutch RMBS	UK cards	European consumer receivable	European auto
	(0.1)	(−0.15)	(−0.63)	(−0.33)	(−0.48)	(−0.91)
$\Delta r-10$	−0.207	−0.140	−0.067	−0.134	−0.106	−0.146
	(−1.36)	(−1.34)	(−0.81)	(−1.84)	(−1.97)	(−1.71)
$(\Delta r-10)^2$	0.194	−0.022	−0.058	−0.036	−0.096	−0.018
	(0.85)	(−0.15)	(−0.5)	(−0.34)	(−1.23)	(−0.15)
ΔS&P	−0.915	−0.272	−0.709	0.164	0.194	0.695
	(−0.76)	(−0.33)	(−0.86)	(0.28)	(0.45)	(1.01)
ΔVIX	−0.006	−0.001	−0.002	0.000	0.001	0.003
	(−0.64)	(−0.11)	(−0.31)	(0.07)	(0.39)	(0.6)
ΔSlope	0.179	0.077	0.051	0.133	0.091	0.190
	(1.05)	(0.66)	(0.51)	(1.64)	(1.53)	(2)
LIB-OIS	25.57	30.71	20.89	14.40	6.56	8.62
F-test	(< 0.01)	(< 0.01)	(< 0.01)	(< 0.01)	(0.01)	(< 0.01)
ABX	0.01	0.28	0.14	0.09	0.51	0.41
F-test	(0.93)	(0.60)	(0.71)	(0.77)	(0.48)	(0.52)

Table 4
Summary of F-test results for different asset categories.

For each bond i, we estimate Eq. (2) using weekly data from January 4, 2007 to January 29, 2009, ΔLIB-OIS is the percentage change of the spread between the 3-month LIBOR and the Overnight Index Swap (OIS). This table summarizes the F-test results for the LIB-OIS state variable and its lags. The null hypothesis of F-test is the sum of all coefficients of ΔLIB-OIS and its lags is zero. The numbers in the table indicate how many F-tests of bonds in each category are significant at various confidence levels. Asset categories are listed in Panel A of Table 1. "Negative" and "Positive" indicate the sign of the sum of coefficients for ΔLIB-OIS and its lags.

Categories	Total number	Negative			Positive		
		10%	5%	1%	10%	5%	1%
Panel A: Whole period: January 4, 2007 to January 29, 2009							
Subprime	63	1	1	0	2	1	2
Nonsubprime_US	176	3	0	0	4	7	106
Nonsubprime_Europe	59	0	0	0	0	6	39
Financial	46	0	0	0	3	2	6
Industrial	48	0	0	0	5	14	9
Total	392	4	1	0	14	30	162
Panel B: Subperiod I: January 4, 2007 to December 27, 2007							
Subprime	63	2	4	2	0	2	1
Nonsubprime_US	176	4	0	1	8	21	75
Nonsubprime_Europe	59	0	0	0	5	10	24
Financial	46	3	1	0	1	2	1
Industrial	48	0	0	0	0	1	4
Total	392	9	5	3	14	36	105
Panel C: Subperiod II: January 3, 2008 to January 29, 2009							
Subprime	63	1	0	0	0	0	0
Nonsubprime_US	176	8	0	0	23	26	41
Nonsubprime_Europe	59	0	0	0	0	1	0
Financial	46	0	0	0	6	10	6
Industrial	48	0	0	0	6	9	21
Total	392	9	0	0	35	46	68

Table 3 continued and Table 4

6.8 Tables 5–7: Rating F-tests, Repo Spreads, and Haircuts

Read: Table 5 shows rating-level evidence. Table 6 shows repo spreads respond to LIB-OIS. Table 7 shows haircuts respond more to collateral volatility.

Economic message. Repo rates price counterparty risk; haircuts are the withdrawal channel.

Table 5
Summary of *F*-test results for different rating classes.

For each bond *i*, we estimate Eq. (2) using weekly data from January 4, 2007 to January 29, 2009, ΔLIB-OIS is the percentage change of the spread between the three-month LIBOR and the overnight index swap (OIS).This table summarizes the *F*-test results for the LIB-OIS state variable and its lags. The null hypothesis of *F*-test is the sum of all coefficients of ΔLIB-OIS and its lags is zero. The numbers in the table indicate how many *F*-tests of bonds in each rating class are significant at various confidence levels. Negative and positive indicate the sign of the sum of the coefficients for ΔLIB-OIS and its lags.

Rating	Total number	Negative			Positive		
		10%	5%	1%	10%	5%	1%
Panel A: Whole Period: January 4, 2007 to January 29, 2009							
AAA	157	4	0	0	4	10	83
AA	47	0	1	0	1	3	9
A	74	0	0	0	3	5	33
BBB	83	0	0	0	2	6	36
Other	31	0	0	0	4	6	1
Total	392	4	1	0	14	30	162
Panel B: Subperiod I: January 4, 2007 to December 27, 2007							
AAA	157	5	1	1	9	25	47
AA	47	0	0	1	0	1	7
A	74	0	2	0	3	4	27
BBB	83	1	1	0	1	5	23
Other	31	3	1	1	1	1	1
Total	392	9	5	3	14	36	105
Panel C: Subperiod II: January 3, 2008 to January 29, 2009							
AAA	157	0	0	0	4	13	44
AA	47	1	0	0	3	12	4
A	74	4	0	0	14	9	5
BBB	83	4	0	0	9	11	5
Other	31	0	0	0	5	1	10
Total	392	9	0	0	35	46	68

(a) Table 5: F-tests by rating

Table 6
Repo spreads regression results.

For each class of securitized bonds, we estimate Eq. (3) using weekly data from January 4, 2007 to January 29, 2009. ΔLIB-OIS is the percentage change of the spread between the 3-month, London Interbank Offered Rate (LIBOR) and the overnight index swap (OIS). ΔABX is the parentage change of the (ABX) index at period *t*. ΔOIS is the Overnight Index Swap. $\Delta r-10$ is the change in yield on the 10-year Treasury, with its square given by $(\Delta r-10)^2$. ΔSlope is the change in 10-year minus 2-year Treasury yields. ΔVIX is the change in implied volatility of S&P 500, and ΔS&P is the return on (S&P) 500. *t*-statistics are given in parentheses below the coefficient estimates. The last two rows report *F*-statistics and *p*-values for the key state variables. The null hypothesis of the LIB-OIS *F*-test is that the sum of all coefficients of ΔLIB-OIS and its lags is zero. The null hypothesis of the ABX *F*-test is the sum of all coefficients of ΔABX and its lags is zero. The null hypothesis of the VOL *F*-test is the sum of all coefficients of VOL and its lags is zero.

	Repo Rate Spreads				
	A-AAA ABSAuto/CC/SL	< AA ABSRMBS/CMBS	AA-AAA ABSRMBS/CMBS	AA-AAA CLO	AA-AAA CDO
Intercept	0.035	0.015	0.016	0.017	0.024
	(0.86)	(0.89)	(0.71)	(0.71)	(0.93)
ΔLIB-OIS	1.321	0.825	1.043	1.025	0.558
	(12)	(17.26)	(16.45)	(15.66)	(7.26)
ΔLIB-OIS, $t-1$	−0.168	−0.004	−0.056	−0.068	0.044
	(−1.58)	(−0.1)	(−0.93)	(−1.08)	(0.67)
ΔLIB-OIS, $t-2$	0.084	0.071	0.099	0.115	0.062
	(0.8)	(1.57)	(1.65)	(1.85)	(0.85)
ΔLIB-OIS, $t-3$	−0.134	0.004	−0.040	−0.021	0.010
	(−1.27)	(0.09)	(−0.66)	(−0.35)	(0.15)
ΔABX	−0.188	−0.152	−0.169	−0.183	−0.031
	(−0.86)	(−1.59)	(−1.32)	(−1.4)	(−0.23)
ΔABX, $t-1$	0.227	0.020	0.072	0.076	0.206
	(1.03)	(0.21)	(0.57)	(0.58)	(1.5)
ΔABX, $t-2$	0.435	0.007	0.092	0.086	0.037
	(1.99)	(0.07)	(0.73)	(0.66)	(0.28)
ΔABX, $t-3$	0.018	0.064	0.057	0.085	0.052
	(0.08)	(0.67)	(0.44)	(0.66)	(0.38)
ΔVOL	−0.002	0.000	0.000	0.000	0.000
	(−0.32)	(0.15)	(0.42)	(0.25)	(0.25)
ΔVOL, $t-1$	0.000	0.000	−0.001	−0.001	−0.002
	(0.03)	(−0.35)	(−0.73)	(−0.6)	(−0.84)
ΔVOL, $t-2$	−0.002	0.000	0.001	0.001	0.001
	(−0.32)	(0.76)	(0.98)	(0.45)	(0.26)
ΔVOL, $t-3$	0.001	0.000	−0.001	−0.001	0.002
	(0.18)	(−0.78)	(−1.19)	(−1.3)	(0.8)
ΔOIS	0.060	−0.044	−0.078	−0.005	0.184
	(0.11)	(−0.19)	(−0.21)	(−0.01)	(0.21)
$\Delta r-10$	0.085	0.023	0.085	0.028	0.142
	(0.4)	(0.27)	(0.73)	(0.23)	(0.79)
$(\Delta r-10)^2$	−0.178	−0.019	0.010	0.019	−0.082
	(−0.52)	(−0.13)	(0.05)	(0.09)	(−0.35)
ΔS&P	−0.374	0.136	−0.158	0.430	−2.373
	(−0.21)	(0.17)	(−0.15)	(0.39)	(−1.38)
ΔVIX	0.004	0.000	0.001	0.006	0.000
	(0.29)	(−0.01)	(0.09)	(0.64)	(−0.01)
ΔSlope	−0.218	−0.080	−0.122	−0.027	−0.196
	(−0.98)	(−0.81)	(−0.92)	(−0.2)	(−0.99)
LIB-OIS *F*-test	32.56	129.61	100.87	96.07	30.37
	(< 0.01)	(< 0.01)	(< 0.01)	(< 0.01)	(< 0.01)
ABX *F*-test	1.44	0.12	0.05	0.07	1.07
	(0.23)	(0.73)	(0.82)	(0.79)	(0.3)
VOL *F*-test	0.53	0.01	0.34	1.80	0.14
	(0.47)	(0.93)	(0.56)	(0.18)	(0.71)

(b) Table 6: repo-spread regressions

Table 7
Haircut regression results.

For each class of securitized bonds, we estimate Eq. (7) using weekly data from January 4, 2007 to January 29, 2009, ΔLIB-OIS is the percentage change of the spread between the three -month London Interbank Offered Rate (LIBOR) and the Overnight Index Swap (OIS). ΔABX is the parentage change of the ABX index at period *t*. ΔOIS is the Overnight Index Swap. $\Delta r-10$ is the change in yield on the ten-year Treasury, with its square given by $(\Delta r-10)^2$. ΔSlope is the change in 10-year minus 2-year Treasury yields. ΔVIX is the change in implied volatility of S&P 500, and ΔS&P is the return on (S&P) 500. *t*-statistics are given in parentheses below the coefficient estimates. The last two rows report *F*-statistics and *p*-values for the key state variables. The null hypothesis of the LIB-OIS *F*-test is that the sum of all coefficients of ΔLIB-OIS and its lags is zero. The null hypothesis of the ABX *F*-test is the sum of all coefficients of ΔABX and its lags is zero. The null hypothesis of the VOL *F*-test is the sum of all coefficients of VOL and its lags is zero.

	Change of haircuts				
	A-AAA ABSAuto/CC/SL	< AA ABSRMBS/CMBS	AA-AAA ABSRMBS/CMBS	AA-AAA CDO	AA-AAA CLO
Intercept	0.00096	0.00266	0.00194	−0.00514	0.00311
	(0.69)	(1.19)	(1.08)	(−0.34)	(1.59)
ΔLIB-OIS	−0.00010	0.00009	0.00010	0.00121	0.00003
	(−1.44)	(0.89)	(1.25)	(0.85)	(0.27)
ΔLIB-OIS, $t-1$	−0.00010	0.00001	−0.00002	0.00079	0.00008
	(−1.53)	(0.07)	(−0.24)	(0.55)	(0.67)
ΔLIB-OIS, $t-2$	0.00005	0.00008	0.00011	−0.00053	−0.00016
	(0.75)	(0.75)	(1.31)	(−0.41)	(−1.45)
ΔLIB-OIS, $t-3$	−0.00001	−0.00014	−0.00010	0.00073	0.00006
	(−0.12)	(−1.2)	(−1.18)	(0.62)	(0.74)
ΔABX	0.00001	0.00000	0.00001	0.00004	0.00000
	(1.05)	(0.33)	(1.19)	(0.69)	(−0.34)
ΔABX, $t-1$	0.00000	−0.00001	0.00000	0.00001	0.00000
	(0.06)	(−1.05)	(−0.64)	(0.2)	(0.18)
ΔABX, $t-2$	0.00000	0.00001	0.00001	0.00002	0.00000
	(0.12)	(0.67)	(0.86)	(0.32)	(0.43)
ΔABX, $t-3$	0.00000	−0.00001	−0.00001	0.00003	−0.00001
	(−0.79)	(−1.4)	(−1.58)	(0.44)	(−1.34)
ΔVOL	0.00036	0.00001	0.00000	0.00311	0.00015
	(2.31)	(0.52)	(−0.06)	(3.19)	(2.05)
ΔVOL, $t-1$	−0.00049	−0.00001	−0.00001	−0.00345	0.00000
	(−2.01)	(−0.24)	(−0.12)	(−2.37)	(0.04)
ΔVOL, $t-2$	0.00049	0.00002	0.00003	0.00235	−0.00016
	(2.09)	(0.36)	(0.34)	(1.57)	(−1.61)
ΔVOL, $t-3$	−0.00005	−0.00001	−0.00004	0.00006	0.00017
	(−0.28)	(−0.52)	(−0.69)	(0.05)	(1.99)
ΔOIS	0.00476	−0.01801	−0.01655	−0.09061	−0.01967
	(0.4)	(−0.92)	(−1.02)	(−0.67)	(−1.2)
$\Delta r-10$	0.00413	−0.01072	−0.00689	0.04610	0.00241
	(0.6)	(−1.02)	(−0.82)	(0.51)	(0.26)
$(\Delta r-10)^2$	0.00703	−0.00617	−0.00882	−0.13209	−0.00100
	(0.64)	(−0.36)	(−0.62)	(−0.97)	(−0.06)
ΔS&P	0.03663	−0.02488	−0.06200	0.07360	−0.00699
	(0.66)	(−0.28)	(−0.89)	(0.09)	(−0.09)
ΔVIX	0.00037	−0.00014	−0.00042	0.00098	−0.00074
	(0.77)	(−0.18)	(−0.68)	(0.14)	(−1.07)
ΔSlope	0.01253	0.01069	0.00682	−0.07066	−0.00906
	(1.63)	(0.85)	(0.67)	(−0.7)	(−0.85)
LIB-OIS *F*-test	1.47	0.06	0.69	0.89	0.01
	(0.23)	(0.81)	(0.41)	(0.35)	(0.91)
ABX *F*-test	0.05	0.58	0.01	0.64	0.32
	(0.83)	(0.45)	(0.92)	(0.43)	(0.57)
VOL *F*-test	4.66	0.07	0.29	5.33	5.53
	(0.03)	(0.80)	(0.60)	(0.02)	(0.02)

Table 7: Haircut regressions

7 Short summary

- The crisis was a run on repo in the securitized-banking system.
- ABX captures subprime fundamentals; LIB-OIS captures counterparty risk.
- LIB-OIS explains broad non-subprime credit spread and repo rate movements.
- Haircuts rise when collateral future value becomes volatile and uncertain.
- The paper links securitization, collateral, repo haircuts, and systemic runs.

8 Conclusion

8.1 Core conclusion

The paper shows that the systemic crisis came from a run on repo, not merely from mortgage losses.

8.2 Economic mechanism

Rising haircuts force deleveraging. Deleveraging lowers asset prices and further weakens collateral, creating a spiral.

8.3 Incremental contribution

The paper's main incremental contribution is the collateral-based account of modern bank runs and the evidence that counterparty-risk stress explains contagion to non-subprime assets.

8.4 Limitations

The analysis is correlational and based on opaque repo-market data, so it does not provide a fully structural causal model.

8.5 Takeaway

Modern banking can run through wholesale funding markets. Deposit insurance alone cannot prevent runs when money-like claims are repo contracts backed by securitized collateral.